

Convergence and Stability for IBVP Discretizations

Recap: $u_t = u_{xx}$ $u(x, t=0) = u_0 = x$
 $u(0, t) = 0$ $u(1, t) = 0$

Discretize in space: $U'(t) = A_h U(t)$

$$A = \frac{1}{h^2} \begin{bmatrix} -2 & 1 & & \\ 1 & -2 & \ddots & \\ & \ddots & \ddots & 1 \\ & & 1 & -2 \end{bmatrix}$$

Then discretize in time:

$$U^{n+1} = R(kA_h) U^n$$

For example, with explicit Euler:

$$U^{n+1} = U^n + kA_h U^n$$

$$U^{n+1} = (I + kA_h) U^n$$

Trapezoidal:

$$U^{n+1} = \underbrace{\left(I - \frac{k}{2} A_h\right)^{-1}}_{R(kA_h)} \underbrace{\left(I + \frac{k}{2} A_h\right)}_{R(kA_h)} U^n$$

Let $B_{k,h} = R(kA_h)$.

So

$$U^{n+1} = B_{k,h} U^n \quad (1)$$

$$\Rightarrow U^N = B_{k,h}^N U^0$$

Let

$$\hat{U}^n = \begin{bmatrix} u(x_1, t_n) \\ u(x_2, t_n) \\ \vdots \\ u(x_m, t_n) \end{bmatrix}$$

Replace $U^n \rightarrow \hat{U}^n$ in (1):

$$\hat{U}^{n+1} = B_{k,h} \hat{U}^n + k\tau^n \quad (2)$$

Subtract (2) - (1):

$$U^{n+1} - \hat{U}^{n+1} = B_{k,h} (U^n - \hat{U}^n) - k\tau^n$$

$$E^{n+1} = B_{k,h} E^n - k\tau^n$$

Let $T = NK$ be fixed.
($N = T/k$)

We want to prove that $\|E^N\| \rightarrow 0$ as $k, h \rightarrow 0$.

$$E^N = B_{k,h}^N E^0 - K \sum_{j=0}^{N-1} B_{k,h}^{N-1-j} \tau^j$$

$$\|E^N\| \leq \|B_{k,h}^N\| \cdot \|E^0\| + K \sum_{j=0}^{N-1} \|B_{k,h}^{N-1-j}\| \cdot \|\tau^j\|$$

$$\leq \|B_{k,h}^N\| \cdot \|E^0\| + \underbrace{KN}_T \underbrace{\|B_{k,h}^{N-1}\|}_{\text{should be } C(T)} \cdot \max_j \|\tau^j\|$$

We now assume:

① Consistency: $\|\tau^j\| \rightarrow 0$ as $k,h \rightarrow 0$
 $\|E^0\| \rightarrow 0$ as $k,h \rightarrow 0$

② $\|B_{k,h}^n\| \leq C(T) \leftarrow \begin{array}{l} \text{indep. of } k,h,n \\ \text{Lax-Richtmeyer} \\ \text{stability} \end{array}$

$$\|E^N\| \leq C(T) \underbrace{(\|E^0\| + T \max_j \|\tau^j\|)}_{\rightarrow 0 \text{ as } k,h \rightarrow 0}$$

So $\|E^N\| \rightarrow 0$ as $h,k \rightarrow 0$.

The Lax-Richtmeyer stability Condition

With Euler's method:

$$B_{k,h} = I + kA_h$$

Take $\|\cdot\| = \|\cdot\|_2$.

$$\|B_{k,h}\|_2 = \max_{\mu \in \sigma(B)} |\mu|$$

$\mu = 1 + K\lambda$ where $\lambda \in \sigma(A_h)$

$$\lambda = \frac{2}{h^2} (\cos(p\pi h) - 1)$$

So $\mu_p = 1 + \frac{2K}{h^2} (\cos(p\pi h) - 1)$

Now $\|B_{k,h}^n\|_2 \leq \|B_{k,h}\|^n = (\max_p |\mu_p|)^n \leq C(T)$

A sufficient condition is
 $|\mu_p| \leq 1$ for all p .

i.e. $-1 \leq 1 + \frac{2K}{h^2} (\cos(p\pi h) - 1) \leq 1$

$$-1 \leq \frac{K}{h^2} (\cos(p\pi h) - 1) \leq 0$$

Now $\cos(p\pi h) \in (-1, 1)$.

So we just need

$$-1 \leq -\frac{2K}{h^2} \Rightarrow K \leq \frac{h^2}{2}$$

$\Rightarrow$ same as the condition
for absolute stability.

Why do we have this additional
condition on K_h for convergence?

Because we need

$KL \sim \frac{K}{h^2}$ to remain bounded.

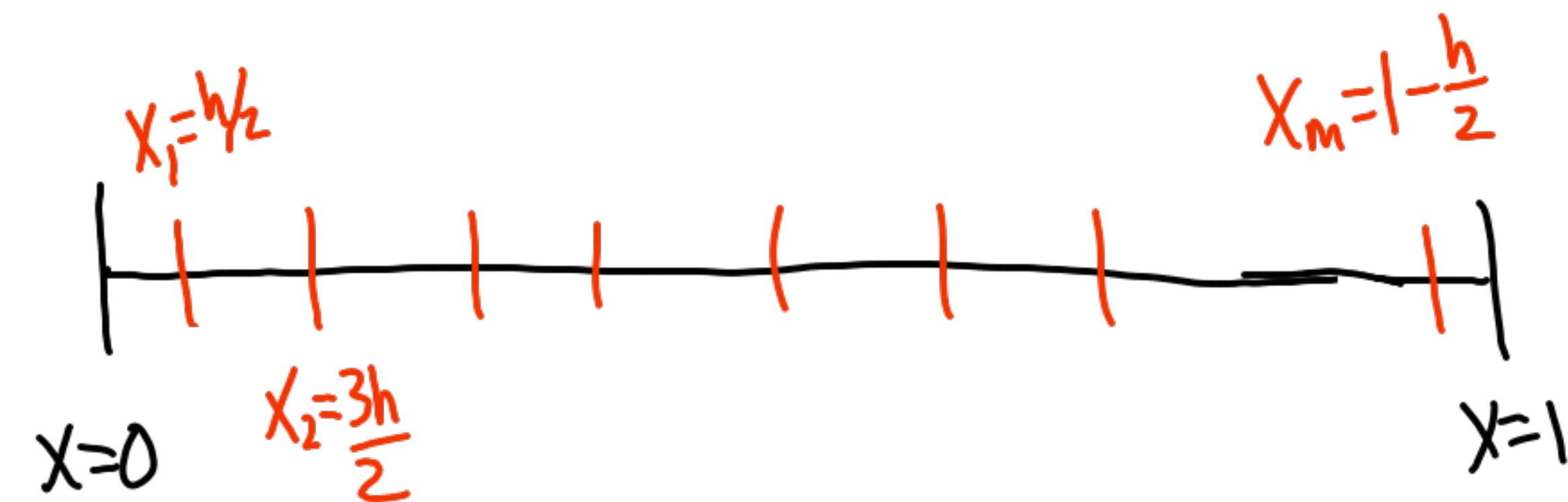
Von Neumann stability analysis

A simple way to analyze stability of linear PDE discretizations.

$$u_t + u_{xx} = 0 \quad 0 < x < 1$$

$$u(x, 0) = \eta$$

$$u(0, t) = u(1, t)$$



$$U_j^{n+1} = U_j^n + \frac{k}{h^2} (U_{j+1}^n - 2U_j^n + U_{j-1}^n)$$

$$2 \leq j \leq m-1$$

$$U_1^{n+1} = U_1^n + \frac{k}{h^2} (U_2^n - 2U_1^n + U_m^n)$$

$$U_m^{n+1} = U_m^n + \frac{k}{h^2} (U_1^n - 2U_m^n + U_{m-1}^n)$$

Ansatz: $U_j^n = g^n e^{ijh\xi}$ $U_{j-1}^n = g^n e^{i(j-1)h\xi}$

$$g^{n+1} e^{ijh\xi} = g^n e^{ijh\xi} + \frac{K}{h^2} g^n e^{ijh\xi} (e^{ih\xi} - 2 + e^{-ih\xi})$$

$$g = 1 + \frac{K}{h^2} (e^{ih\xi} + e^{-ih\xi} - 2)$$

$$= 1 + \frac{K}{h^2} (2\cos(h\xi) - 2)$$

$$g = 1 + \frac{2K}{h^2} (\cos(h\xi) - 1)$$

Stable if $|g| \leq 1 \Rightarrow K \leq \frac{h^2}{2}$

Why do we get the same expression?

We have $U^{n+1} = BU^n$

Where $B = I + K\hat{A}$

$$\hat{A} = \begin{bmatrix} -2 & 1 & & & \\ & 1 & \ddots & & \\ & & \ddots & \ddots & \\ & & & \ddots & 1 \\ 1 & & & & -2 \end{bmatrix}$$

Circulant Matrix

Every FD discretization of a linear PDE yields a circulant matrix.

Every circulant matrix has the same eigenvectors!

$$V_j = e^{ijh}$$

VN analysis takes advantage of this to compute the eigenvalues.

Typically VN analysis provides necessary but not sufficient conditions for stability under other BCs.